

Puzzle-like molecular assembly of non-flammable solid-state polymer electrolytes for safe and high-voltage lithium metal batteries

Received: 19 September 2024

Accepted: 19 August 2025

Published online: 26 September 2025

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Developing safe and high-voltage solid-state polymer electrolytes for high-specific-energy lithium metal batteries holds great promise. However, low ionic conductivity, limited Li^+ transference number, narrow voltage window, and high flammability greatly hinder their practical applications. Herein, we propose a puzzle-like molecular assembly strategy to construct a solid-state polymer electrolyte via in situ polymerization. The triallyl phosphate and 2,2,3,3,4,4,4-heptafluorobutyl methacrylate segments are spliced into the vinyl ethylene carbonate matrix to enhance anion affinity and promote lithium salt dissociation, resulting in a high ionic conductivity of 0.432 mS cm^{-1} and a Li^+ transference number of 0.70 at 25°C . Meanwhile, the polymer electrolyte exhibits a high oxidation voltage of 5.15 V, enabled by its intrinsic high-voltage tolerance and the formation of a robust inorganic-rich interphase. As a result, the $\text{Li}||\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ cell maintains stable performance for 300 cycles and reliably cycles even with an application-oriented mass loading of 15.8 mg cm^{-2} . The 2.6-Ah $\text{Li}||\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ pouch cell reaches a high specific energy of 349 Wh kg^{-1} . Furthermore, the developed polymer electrolyte displays superior nonflammability and the $\text{Li}||\text{LiFePO}_4$ cell exhibits stable cycling for over 120 cycles at 100°C . Both accelerating rate calorimetry and nail penetration tests verify the high safety of the pouch cells using the designed polymer electrolyte, showing the potential for practical applications.

One of the most significant concerns of commercial liquid electrolyte-based lithium-ion batteries (LIBs) is their flammability and explosiveness, greatly hindering their development. Additionally, the utilization of graphite-based negative electrode limits the specific energy of LIBs to typically below 250 Wh kg^{-1} , falling short of the increasing energy demands, particularly in terms of the range for electric vehicles^{1,2}. Lithium metal, with its high theoretical specific capacity

($\sim 3860 \text{ mAh g}^{-1}$) and lowest reduction potential ($\sim 3.04 \text{ V}$ vs. the standard hydrogen electrode), is considered the holy grail for the negative electrode in lithium batteries. Moreover, pairing it with high-voltage positive electrodes can greatly enhance the specific energy of the batteries^{3,4}. Unfortunately, current commercial liquid electrolytes are difficult to be compatible with both lithium metal anodes (LMAs) and high-voltage positive electrodes⁴. Therefore, it is significant to develop

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advanced electrolytes that ensure high safety and compatibility with both LMAs and high-voltage positive electrodes.

Solid-state polymer electrolytes (SPEs) offer promising solutions to the aforementioned challenges due to their advantages, such as the reduced risk of liquid leakage, high flexibility, lightweight properties, and ease of processability^{1,5}. In particular, SPEs can establish intimate contact between electrodes and electrolytes via in situ polymerization, thereby reducing interfacial resistance. Moreover, the in situ polymerization strategy is compatible with existing liquid-based electrolyte battery industry systems and holds the potential for large-scale production of solid-state lithium metal batteries (SSLMBs)^{6,7}. However, many polymer electrolytes still face significant challenges that impede their development. These challenges include low ionic conductivity at 25 °C, low Li^+ transference number (t_{Li^+}), poor high-voltage performance, instability with LMAs, and high flammability^{8,9}. Consequently, extensive research has been dedicated to modifying SPEs to overcome these limitations.

Among SPEs, polycarbonate-based polymer electrolytes have garnered significant attention due to their advantages, such as reasonable coordination strength with Li^+ and compatibility with high-voltage positive electrodes^{10–12}. However, these polymer electrolytes encounter several challenges, including difficulties in lithium salt dissociation and low t_{Li^+} . Additionally, they exhibit instability towards LMAs, leading to severe side reactions^{11,12}. Unmodified polycarbonate-based solid-state electrolytes, despite mitigating the risk of organic electrolyte leakage, remain flammable, necessitating enhanced safety performances for batteries. Although various target-oriented strategies have been proposed to overcome these disadvantages, including incorporating inorganic fillers^{13–15}, blending organic materials^{16,17}, editing molecular structures^{18,19}, forming gel electrolytes by combining with liquid electrolytes²⁰ and designing new salts²¹, these strategies possess somewhat limitations in practical applications. They still fail to simultaneously achieve excellent lithium salt dissociation capability, high t_{Li^+} , stable LMA performance, high-voltage stability, and superior flame retardancy. Therefore, a facile and scalable preparation strategy is urgently desired to enhance the overall performance of polycarbonate-based solid-state electrolytes.

In this work, we employ a puzzle-like molecular assembly approach to design a new SPE. This method involves piecing together three distinct molecular fragments, each representing specific functionalities, through a facile and scalable in situ polymerization strategy. The resulting SPE (named IWSWN-SPE) exhibits improved Li salt dissociation, a wide electrochemical stability window (up to 5.15 V), stable electrolyte/electrode interfaces, a wide temperature range (up to 100 °C), and nonflammability. The carbonate ester (vinyl ethylene carbonate, VEC) fragments help to conduct Li^+ ions, while the introduction of fluorine- (F-) (2,2,3,3,4,4,4-heptafluorobutyl methacrylate, HFBMA) and phosphorus-rich (P-rich) (triallyl phosphate, TAP) fragments can facilitate the dissociation of lithium bis(trifluoromethane) sulfonimide (LiTFSI) and restrict the migration of TFSI^- . As a result, the t_{Li^+} is increased to 0.70. Moreover, incorporating P-rich segments endows the superior fire-retardant feature of IWSWN-SPE, thereby ensuring the high safety of the high-voltage Li metal batteries (LMBs). The assembled $\text{Li}|\text{IWSWN-SPE}|\text{LFP}$ cell can stably for 120 cycles even at 100 °C. Additionally, the incorporation of F-rich and P-rich molecular segments imparts exceptional electrochemical stability to IWSWN-SPE when subjected to the high-voltage positive electrode and LMA, thereby guaranteeing the outstanding electrochemical performance of LMBs. The $\text{Li}|\text{IWSWN-SPE}|\text{Li}$ symmetric cell shows excellent Li plating/stripping stability, maintaining for over 2500 h at 0.2 mA cm^{-2} . The $\text{Li}|\text{IWSWN-SPE}|\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2$ (NCM622) cell can stably for 300 cycles and reliably cycle for over 100 cycles even with a high mass loading of 15.8 mg cm^{-2} at 30 °C. In addition, the assembled 2.6-Ah solid-state $\text{Li}|\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM811) pouch cell can achieve a high specific energy of 349 Wh kg^{-1} (based on the 28.2 g of the pouch

cell). Furthermore, our IWSWN-SPE have demonstrated superior safety performance through accelerated rate calorimetry (ARC) testing of pouch cells and nail penetration experiments on Ah-level pouch cells.

Results and discussion

The preparation of IWSWN-SPE

As illustrated in Fig. 1a, we utilize a puzzle-like assembly strategy, employing free radical polymerization of double bonds to combine three molecule segments with specific functionalities (Fig. S1, supporting information) into an SPE. Regarding the concept of puzzle-like assembly, we use the analogy of assembling puzzles by interlocking puzzle pieces to illustrate how we design and synthesize the SPE at the molecular level. In our work, we are not attempting to precisely control the self-assembly of the monomers through free radical polymerization. The monomers can polymerize with themselves or with other different monomers (Fig. S2 and Supplementary Note 1).

The VEC molecule segment interacts with Li^+ and aids in Li^+ conduction^{22,23}. However, PVEC (polymer solely polymerized from VEC monomers) suffers from drawbacks such as flammability and instability with Li metal⁷. The F-rich HFBMA molecule segment with a low highest occupied molecular orbital (HOMO) level can withstand high voltage and contribute to forming a robust solid electrolyte interphase (SEI) and cathode-electrolyte interphase (CEI), making it suitable for LMA, but it suffers from weakened Li^+ dissociation and flammability²⁴. The P-containing TAP molecule segment offers favorable stability at high voltage and good flame retardancy due to its superior ability to trap $\text{H}^\cdot$ and $\text{OH}^\cdot$, but it suffers from severe side reactions with LMA^{25,26}. After the three segments with specific functionalities are molecularly spliced together through a facile in situ free radical polymerization, a high-performance SPE with favorable Li^+ conductivity, good nonflammability, high-voltage stability, and compatibility towards LMA is obtained.

The polymerization of the IWSWN-SPE is confirmed using the ^{13}C nuclear magnetic resonance (NMR) and the solid-state nuclear magnetic resonance (ssNMR, ^{19}F and ^{31}P), as the spectra are shown in Figs. S3, 1b and 1c. The signals of ^{19}F ssNMR spectra (Fig. 1b) indicate the incorporation of the HFBMA segment. As shown in the ^{31}P ssNMR spectrum (Fig. 1c), the chemical shift around 0 ppm is attributed to the $-\text{PO}_4$ group, and the peak shape is broad, which corresponds to the characteristic peak of the polymeric TAP. Moreover, we employ thermogravimetric analysis (TGA) analysis to determine the residual VEC content, as depicted in Fig. 1d. The TGA results indicated a residual VEC content of 54% for PVEC and 23% for our IWSWN-SPE. As shown in Figs. S4 and S5, in contrast to the gel-like state of PVEC (Fig. S4), the in situ polymerized IWSWN-SPE exhibits a dry state (Fig. S5). The Fourier transform infrared spectra (FTIR) are further conducted to test the chemical structure of IWSWN-SPE (powder obtained after grinding and vacuum drying, Fig. 1e). The signals at 1782/1052, 1734/1048, and 1778/1056 cm^{-1} in the spectra of VEC, HFBMA, and IWSWN-SPE belong to the stretching vibration of $\text{C}=\text{O}/\text{C}-\text{O}-\text{C}$ groups. Also, the peak belonging to $\text{C}=\text{C}$ is negligible in IWSWN-SPE, indicating that polymerization has occurred. The existence of F and P is also confirmed based on the X-ray photoelectron spectroscopy (XPS) spectra (Fig. S6).

Al_2O_3 -coated polyethylene separator is used as the scaffold for reinforcing the mechanical properties of IWSWN-SPE. After polymerization, a flat and uniform electrolyte surface morphology is exhibited (Fig. S7). The microstructure of the NCM622 positive electrode before and after polymerization is displayed in Fig. S8. Compared with the pristine NCM622 positive electrode, the positive electrode particles are uniformly embedded in the polymer electrolyte after in situ polymerization, which is instrumental in realizing the rapid migration of Li ions in the electrode. Atomic force microscopy (AFM) is further performed to characterize the surface smoothness of the prepared polymer film (Fig. 1f). The result shows the maximum height difference is only 50 nm, confirming that the surface is smooth,

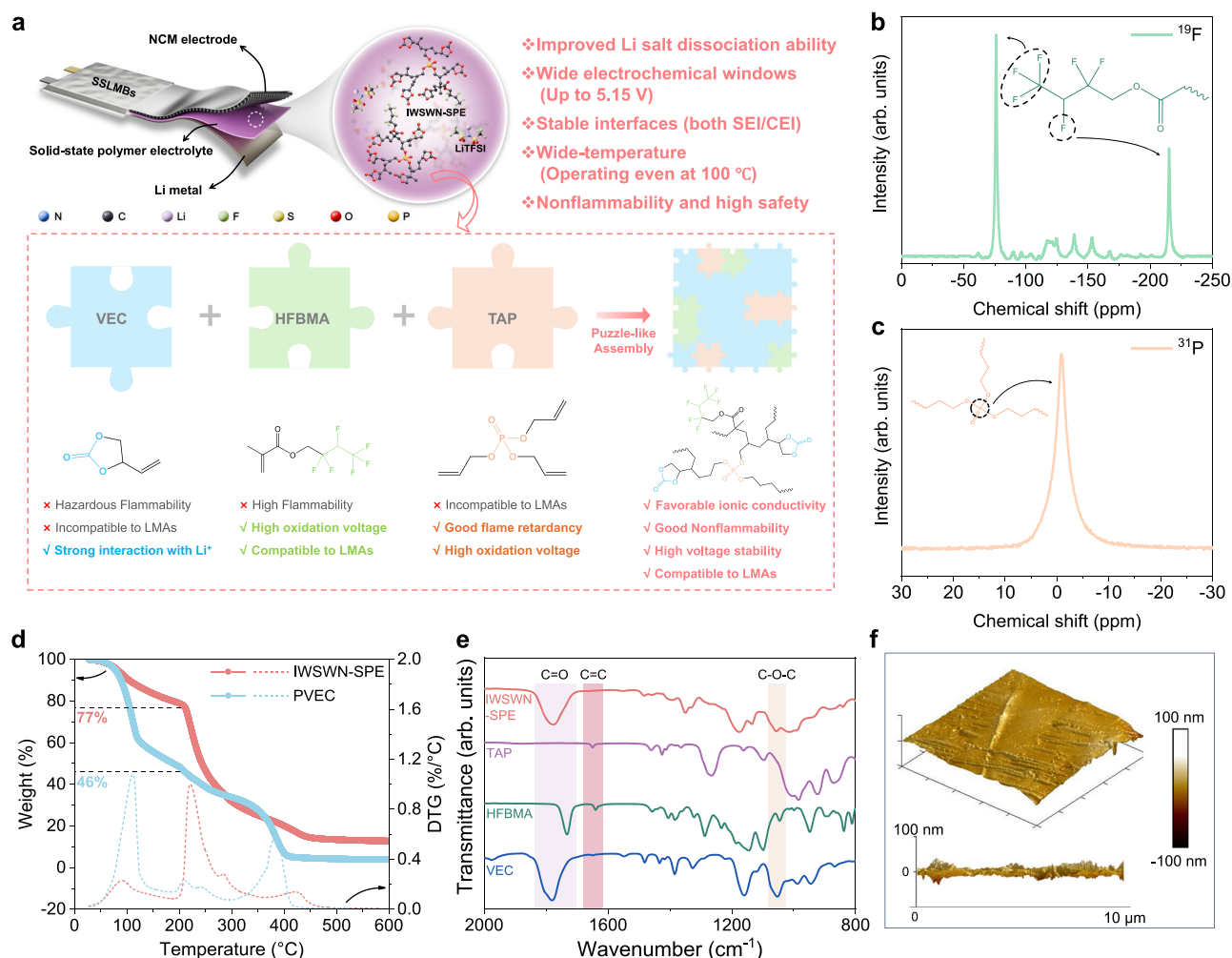


Fig. 1 | Synthetic route for IWSWN-SPE and structure characterizations. **a** Designed route of the IWSWN-SPE. **b, c** The ^{19}F and ^{31}P ssNMR spectra of the IWSWN-SPE. **d** TGA curves of PVEC and IWSWN-SPE. **e** FTIR spectra of the VEC, HFBMA, TAP, and IWSWN-SPE. **f** AFM images showing the smoothness of the IWSWN-SPE film.

consistent with the results obtained from scanning electron micrographs (SEM). This indicates that the precursors of three molecule segments are uniformly dispersed, which is conducive to each component fully exerting its function and achieving the targeted performance.

Physicochemical/ electrochemical properties of the IWSWN-SPE

Through a series of screening experiments, the molar ratio of each monomer is determined to be $n(\text{VEC}):n(\text{HFBMA}):n(\text{TAP}) = 1:0.1:0.15$ (Figs. S9–S11 and Supplementary Note 2). The amount of LiTFSI (10–40%) is optimized to achieve the highest ionic conductivity of the IWSWN-SPE. When the mass ratio of LiTFSI is 20%, the ionic conductivity at 25 °C reaches $4.32 \times 10^{-4} \text{ S cm}^{-1}$ (Fig. 2a and Fig. S12). Specifically, the ionic conductivity of PVEC at 25 °C is $5.97 \times 10^{-4} \text{ S cm}^{-1}$ (Fig. S13), which can be attributed to its 54% content of unreacted VEC monomers. t_{Li^+} is another critical parameter for SPEs, as a low t_{Li^+} leads to the formation of a space charge layer and increased electrode polarization^{27,28}. The t_{Li^+} of IWSWN-SPE and PVEC is measured through chronoamperometry in combination with electrochemical impedance spectroscopy (Fig. S14). In contrast to the PVEC, the t_{Li^+} of the IWSWN-SPE is significantly increased from 0.36 to 0.70 after introducing the F- and P-containing molecular fragments. Compared to various polymer-based electrolytes reported in recent literatures^{27,29–43} (Fig. 2b and Table S1), IWSWN-SPE exhibits one of the highest t_{Li^+} values.

Linear sweeping voltammetry (LSV) tests are conducted to assess the electrochemical stabilities of the PVEC and IWSWN-SPE (Fig. 2c).

The introduction of HFBMA and TAP molecule segments in the IWSWN-SPE increases the oxidation potential from 4.37 V to 5.15 V, indicating the suitability of the IWSWN-SPE for high-voltage positive electrodes. Furthermore, the oxidation stability of IWSWN-SPE is further validated using an electrochemical floating test (Fig. 2d)¹. When the voltage reaches 4.5 V, the leakage current of the PVEC dramatically increases. However, for the IWSWN-SPE, even at a high voltage of 4.9 V, the leakage current remains low ($\sim 5 \mu\text{A}$) and stable.

DFT calculations are conducted to evaluate the high-voltage stability of IWSWN-SPE (Fig. 2e). For the IWSWN-SPE system, given that it is a polymer with a repeating unit structure, we conducted the calculations on the representative repeating unit, as shown in Fig. S15. As depicted in Fig. 2e, the IWSWN-SPE exhibits the lowest HOMO energy level of -8.04 eV , indicating strong oxidative stability. The HOMO orbital distribution primarily resides on the HFBMA-based and TAP-based moieties, suggesting that these segments play a key role in oxidation reactions. Furthermore, the IWSWN-SPE demonstrates a relatively high lowest unoccupied molecular orbital (LUMO) energy level of -0.45 eV , which represents a significant improvement in reductive stability compared to VEC (-0.83 eV). For IWSWN-SPE, the LUMO orbital distribution is predominantly localized on the HFBMA-based fragment, further supporting our hypothesis that this segment undergoes preferential reduction to yield an SEI with a high fluorine content.

The simulations of the electron cloud density distribution of VEC, HFBMA, TAP, and IWSWN-SPE units are performed and are shown in

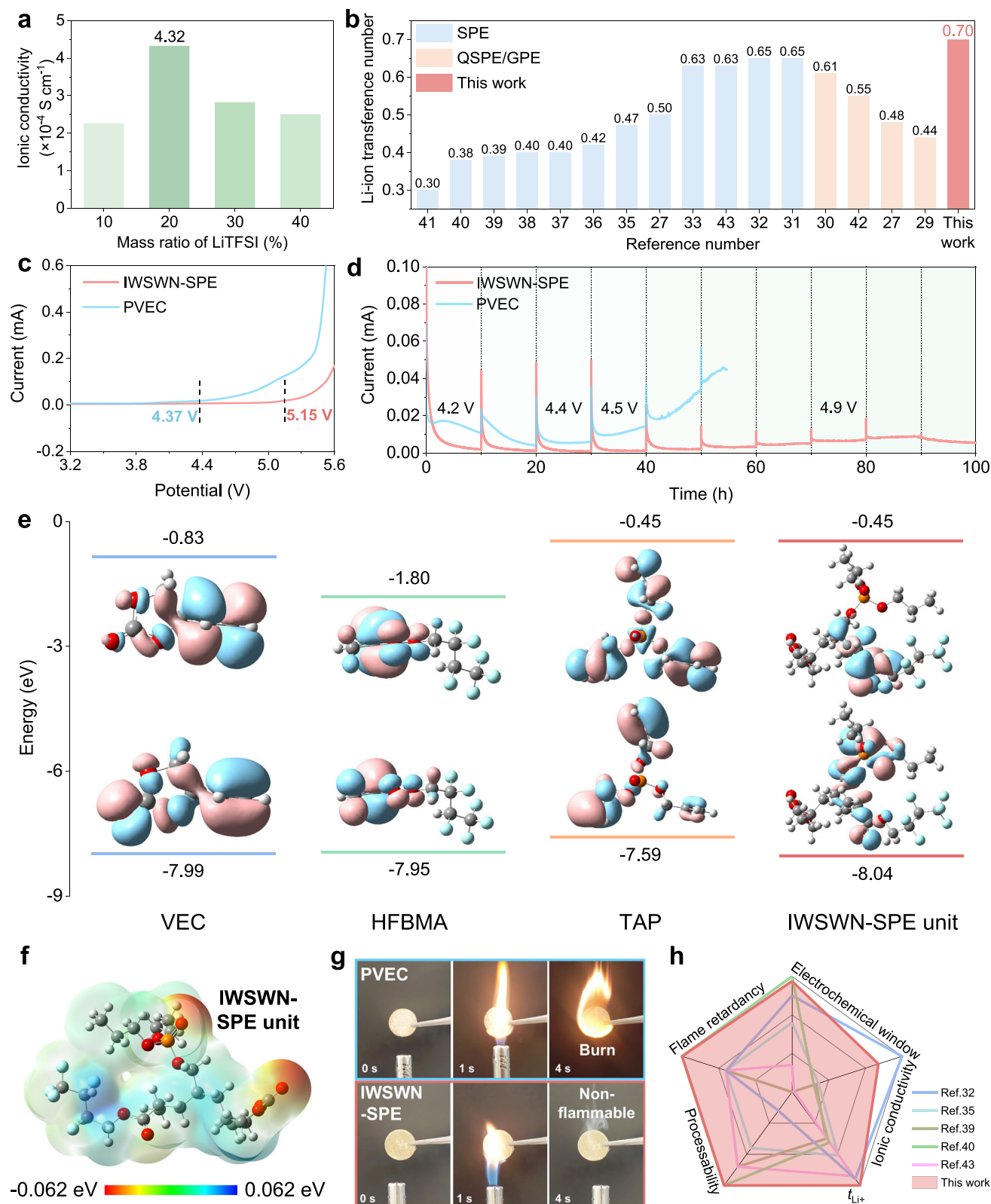


Fig. 2 | The properties of the IWSWN-SPE. a Ionic conductivity of IWSWN-SPE with different mass ratios of LiTFSI at 25 °C. **b** Comparison of t_{Li^+} of different polymer-based electrolytes. **c** LSV test results for IWSWN-SPE and PVEC (scan rate: 1 mV s^{-1}). **d** Electrochemical floating experiments of the Li||NCM622 cells with different polymer electrolytes at the voltage range of 4.2–5.1 V with an increment of 0.1 V.

e Calculated HOMO/LUMO energy levels of VEC, HFBMA, TAP, and IWSWN-SPE unit. **f** Calculated simulations on the electron cloud density distribution of IWSWN-SPE. **g** Burning tests of PVEC and IWSWN-SPE. **h** Radar plots comparing our IWSWN-SPE to other reported polymer-based electrolytes.

Fig. 2f and Fig. S16. For the IWSWN-SPE, the negative electrostatic potential regions (electron-donor sites) are primarily localized on the $O_{C=O}$ in the VEC-based segment and the $O_{P=O}$ in the TAP-based segment. These regions are known to preferentially interact with Li^+ , facilitating Li^+ coordination and transport within the polymer matrix. Conversely, the positive electrostatic potential regions are mainly concentrated on the H_{CF-H} in the HFBMA-based segment. These positively charged H atoms are more likely to interact with TFSI, providing potential anion-binding sites.

Superior nonflammability is one of the key advantages of our designed IWSWN-SPE. As shown in Fig. 2g, the PVEC ignites vigorously and continues to burn even after removing the flame, indicating that PVEC is highly flammable. In contrast, IWSWN-SPE remains non-flammable even when exposed to a flame and retains its original shape after the flame is extinguished. This demonstrates that our IWSWN-SPE is highly flame retarded, which is a crucial enhancement for the safety performance of LMBs. The overall performance properties of IWSWN-SPE and other reported polymer-based solid electrolytes are summarized in radar plots (Fig. 2h and Table S2)^{32,35,39,40,43}. Our IWSWN-SPE exhibits a wide electrochemical window, high ionic conductivity, high Li^+ transference number, excellent processability, and good flame retardancy.

The environment of Li^+ in the IWSWN-SPE system

The temperature-dependence ionic conductivity of PVEC and IWSWN-SPE is investigated across the temperature range of 25 to 80 °C (Fig. 3a). The calculated activation energy (E_a) of IWSWN-SPE (0.387 eV) is smaller than that of PVEC (0.416 eV), indicating that the IWSWN-SPE has a lower energy barrier for Li^+ conduction. The Raman shift between 720 and 760 cm^{-1} is ascribed to the coupled $-CF_3$ bending and S–N stretching of TFSI. An obvious shift of the IWSWN-SPE toward a lower wavenumber (Fig. 3b)⁴⁴ indicates that $LiTFSI$ is highly dissociated in the IWSWN-SPE (higher than PVEC), which is a prerequisite for promoting ion conduction. Furthermore, the 7Li ssNMR test is conducted to identify the interaction and the variation of the chemical environment around the Li^+ in the IWSWN-SPE system (Fig. 3c). Compared with pure $LiTFSI$, the 7Li resonance of IWSWN-SPE shifts to a higher frequency (from -0.60 ppm to -0.45 ppm), corresponding to the increased mobility of Li^+ ²⁷. These results collectively indicate that in the IWSWN-SPE, $LiTFSI$ is more prone to dissociation compared to Li salt in PVEC. Furthermore, the theoretical calculations reveal that the Li^+ dissociation energy in the IWSWN-SPE (0.76 eV) is significantly lower than that required in PVEC (2.97 eV), further suggesting that the dissociation of $LiTFSI$ is more favorable in the IWSWN-SPE (Fig. 3d).

The Li^+ transfer mechanisms in IWSWN-SPE are elucidated using molecular dynamics (MD) simulations, and the interactions between $Li^+/TFSI^-$ ions and PVEC/IWSWN-SPE are studied. The key thermodynamic and structural parameters, including total energy, system volume, density, and temperature, show no significant fluctuations after 50 ns, indicating that both systems had reached equilibrium (Fig. S17). The molecules at the energy equilibrium state and the snapshots evolve from MD simulations of PVEC and IWSWN-SPE systems are shown in Figs. 3e, f, S18, and S19. In both systems, Li^+ and $TFSI^-$ are surrounded by the polymer chain and VEC monomer. The radial distribution function (RDF) and coordination number (CN) analyses are performed to investigate the coordination environment of Li^+ in both PVEC and IWSWN-SPE systems (Fig. 3g, h). The specific value of the coordination number is depicted in Fig. 3i. For the $Li-O$ ($TFSI^-$) coordination, $CN(Li-O_{TFSI^-})$ of the PVEC system is 0.41 and $CN(Li-O_{TFSI^-})$ of the IWSWN-SPE system is 0.21. A lower $Li-O_{TFSI^-}$ coordination in IWSWN-SPE indicates weaker Li^+-TFSI^- association compared to PVEC. Li^+ coordination with oxygen atoms of the polymer chain is more pronounced in the IWSWN-SPE system ($Li-O_{C/P=O}$: 1.28) than in PVEC ($Li-O_{C=O}$: 0.95), suggesting that Li^+ is more likely coordinated with the polymer backbone in IWSWN-SPE. Moreover, the ionic conductivity of

PVEC is not lower than that of IWSWN-SPE. This can be attributed to the presence of unpolymerized VEC monomers in PVEC, which enhances Li^+ coordination and mobility. RDF analysis further revealed that VEC monomers play a significant role in Li^+ transport in PVEC, as evidenced by the higher CN of Li^+ with $O_{C=O}$ atoms from VEC (2.73 in the PVEC system vs. 2.53 in the IWSWN-SPE system).

To avoid the effect of VEC content on conductivity, we subjected PVEC to vacuum-heating and determined, from the mass loss, that its VEC content could be reduced to ~25%. The ionic conductivity measurements on the (25%-VEC) PVEC yielded $2.12 \times 10^{-4} S cm^{-1}$, which is lower than the conductivity of $4.35 \times 10^{-4} S cm^{-1}$ observed for the 23%-VEC IWSWN-SPE (Fig. S20). This is owing to distinct coordination environments for Li^+ in the two systems, which can be observed from the MD simulation snapshots. In the PVEC system, Li^+ primarily coordinates with the $C=O$ groups on the PVEC main chain and the VEC monomer (Fig. 3j). In contrast, in the IWSWN-SPE system, Li^+ not only coordinates with the $C=O$ groups on the IWSWN-SPE main chain and the VEC monomer but also interacts with the $P=O$ groups on the TAP fragment of the IWSWN-SPE backbone (Fig. 3k). This introduces a more diverse coordination environment for Li^+ in the IWSWN-SPE system, offering multiple pathways for Li^+ transport. More importantly, we assess the binding energy (BE) of Li^+ with the polymer systems (as shown in Fig. S21). The BE between PVEC and Li^+ is calculated to be -3.75 eV. In contrast, our IWSWN-SPE shows a weaker BE of -3.40 eV. This reduced binding strength is crucial for Li^+ transport. A weaker interaction implies that the Li^+ is less tightly bound to the polymer chains, facilitating a more labile solvation environment^{45,46}. Consequently, the energy barrier for Li^+ to dissociate from one coordination site and hop to an adjacent one is lowered. This aligns with the principle that overly strong cation-dipole interactions can hinder ion mobility. For IWSWN-SPE, we effectively reduce these interactions, providing a moderate environment for Li^+ .

The diffusion coefficients of Li^+ and $TFSI^-$ are analyzed in Fig. S22 and Supplementary Note 3. It shows that the $TFSI^-$ diffusion coefficient in the PVEC system ($8.64 \times 10^{-8} cm^2 s^{-1}$) is 2.67 times higher than in the IWSWN-SPE system ($3.24 \times 10^{-8} cm^2 s^{-1}$), while Li^+ diffusion coefficients in the two systems (3.73×10^{-8} vs. $3.51 \times 10^{-8} cm^2 s^{-1}$) remain similar. The reduced mobility of $TFSI^-$ in IWSWN-SPE is attributed to stronger interactions between $TFSI^-$ and the polymer chain of IWSWN-SPE, as confirmed by the following: 1) Structural snapshots from the simulations (Fig. 3l, m) further confirm this observation. In the PVEC system, the distances between the N atom of $TFSI^-$ and the oxygen atoms of the nearest polymer chain are 4.42 Å and 5.16 Å. In contrast, in the IWSWN-SPE system, these distances are reduced to 3.98 Å and 4.06 Å, indicating stronger interactions. 2) Additionally, BE of $TFSI^-$ with PVEC/IWSWN-SPE (Fig. 3n) calculations further support this observation. The BE of PVEC- $TFSI^-$ is -1.51 eV, which is more positive than the BE of IWSWN-SPE- $TFSI^-$ (-2.11 eV), indicating that $TFSI^-$ has a higher attractive interaction with the IWSWN-SPE than the PVEC. Moreover, the introduction of TAP and HFBMA fragments makes it easier for the IWSWN-SPE to bind with $TFSI^-$ anions (Fig. S23), which is consistent with the analysis in Fig. 2f. This reduced $TFSI^-$ mobility is a critical factor contributing to the higher t_{Li^+} observed in the IWSWN-SPE system compared to the PVEC system. In the IWSWN-SPE system, the dissociated $TFSI^-$ is more inclined to attach to the main chain of the polymer compared to the PVEC system, restricting the free movement of $TFSI^-$, thereby contributing to a higher t_{Li^+} .

The electrochemical performance of batteries with the IWSWN-SPE

The Li plating/stripping performances of $Li|SPEs|Li$ symmetric cells are tested to investigate the interfacial compatibility and reversibility of the LMAs. As shown in Fig. 4a, a short-circuit occurs in the cell using the PVEC after 615 h at $0.2 mA cm^{-2}$ with an area capacity of $0.2 mAh cm^{-2}$. In contrast, the symmetric cell with the IWSWN-SPE can

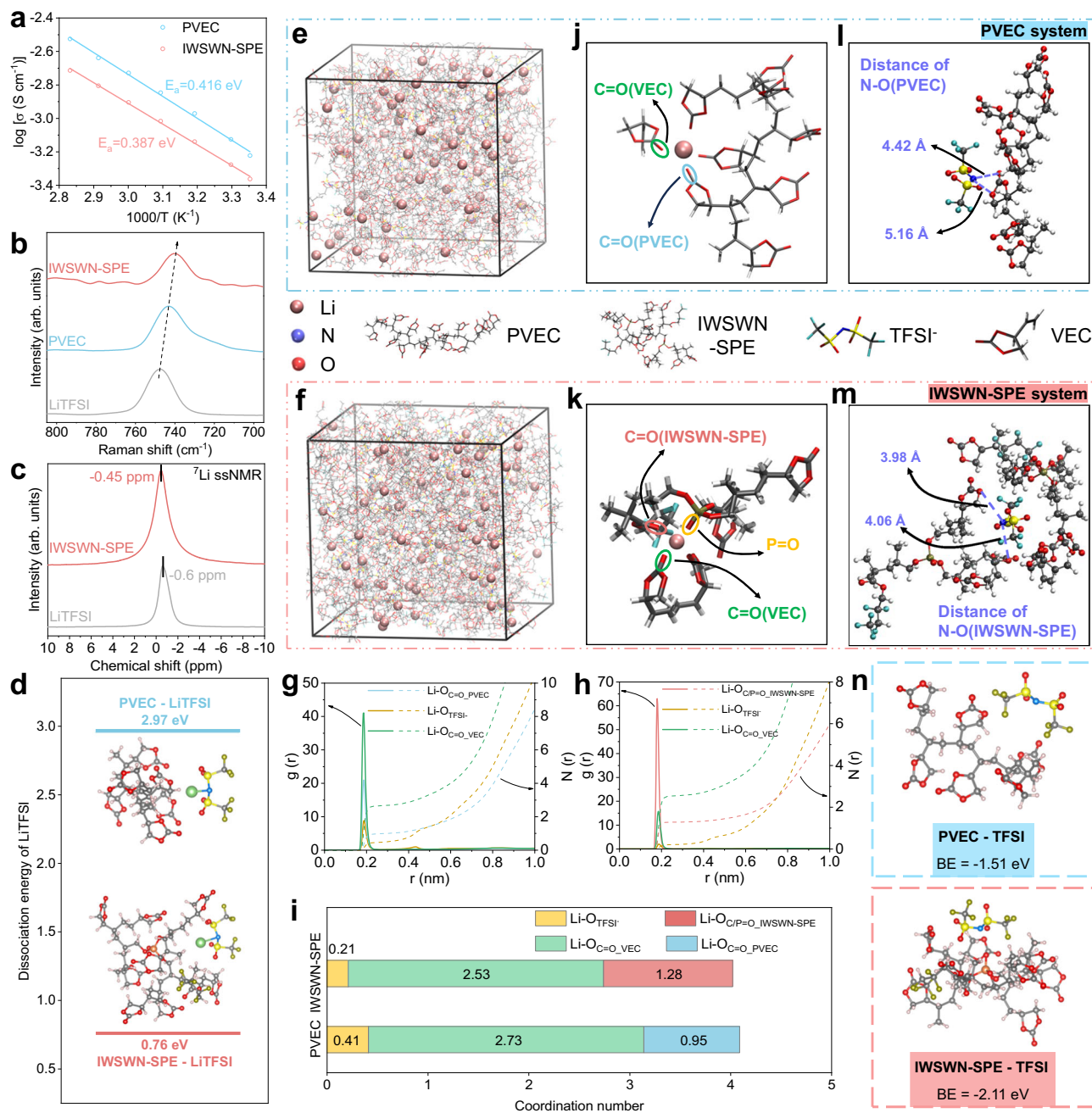


Fig. 3 | The environment of Li^+ in the PVEC and IWSWN-SPE system. **a** Ionic conductivity dependence with varied temperatures from 25°C to 80°C for stainless steel (SS)|PVEC/IWSWN-SPE|SS symmetrical cells. **b** Raman spectra in the TFSI band. **c** ^7Li ssNMR spectra comparison of IWSWN-SPE and LiTFSI. **d** Calculation results of the dissociation energy of LiTFSI in PVEC and IWSWN-SPE systems. (Li: green; N: blue; S: yellow; O: red; P: orange; C: gray; H: pink; F: dark yellow). **e, f** 3D snapshot of the environment centered on a Li^+ within the simulation model for PVEC (**e**) and IWSWN-SPE (**f**) systems. **g, h** RDF and CN of the PVEC (**g**) and IWSWN-SPE (**h**) systems. **i** The comparison results of the CN in PVEC and IWSWN-SPE systems. **j, k** The coordinated environment of Li^+ in the PVEC (**j**) and IWSWN-SPE (**k**) systems. **l, m** Distance measurement graph between the N atom of TFSI and the O atoms of the nearest polymer chain in the PVEC (**l**) and IWSWN-SPE (**m**) systems. (Li: rose red; N: blue; S: yellow; O: red; P: orange; C: gray; H: pink; F: dark yellow). **n** Calculated results of the binding energy (BE) of the PVEC-TFSI and IWSWN-SPE-TFSI. (N: blue; S: yellow; O: red; P: orange; C: gray; H: pink; F: dark yellow).

stably operate for more than 2600 h without experiencing a short circuit or notable overpotential increase. Also, the Li||IWSWN-SPE||Li cell has low and stable overpotentials of 45, 88, 145, and 413 mV for 600 h of consecutive Li plating/stripping at current densities of 0.1, 0.2, 0.3, and 0.5 mA cm^{-2} , respectively (Fig. 4b and Fig. S24). Meanwhile, the rate performance test with a step-increase current density from 0.1 to 1.5 mA cm^{-2} is also performed to further evaluate the critical current density (CCD) of IWSWN-SPE (Fig. 4c). The IWSWN-SPE shows a CCD up to 1.3 mA cm^{-2} , indicating a high current tolerance.

SPE (**h**) systems. **i** The comparison results of the CN in PVEC and IWSWN-SPE systems. **j, k** The coordinated environment of Li^+ in the PVEC (**j**) and IWSWN-SPE (**k**) systems. **l, m** Distance measurement graph between the N atom of TFSI and the O atoms of the nearest polymer chain in the PVEC (**l**) and IWSWN-SPE (**m**) systems. (Li: rose red; N: blue; S: yellow; O: red; P: orange; C: gray; H: pink; F: dark yellow). **n** Calculated results of the binding energy (BE) of the PVEC-TFSI and IWSWN-SPE-TFSI. (N: blue; S: yellow; O: red; P: orange; C: gray; H: pink; F: dark yellow).

The IWSWN-SPE with wide electrochemical windows is suitable for rechargeable LMBs with high-voltage NCM622 positive electrodes. Figs. 4d, e show the rate capabilities and charge/discharge curves of the Li||IWSWN-SPE||NCM622 ($1 \text{ C} = 180 \text{ mA g}^{-1}$), indicating reversible capacity values of 173.9, 168.0, 161.8, 149.6, and $118.4 \text{ mA h g}^{-1}$ at 0.1, 0.2, 0.3, 0.5, and 1.0 C , respectively. These values are much higher than those obtained for the Li|PVEC|NCM622 cell under various rates (Figs. 4d, f). Fig. 4g shows the cycling performance of the Li||NCM622 cell with different polymer electrolytes within a voltage range of 2.5–4.3 V under

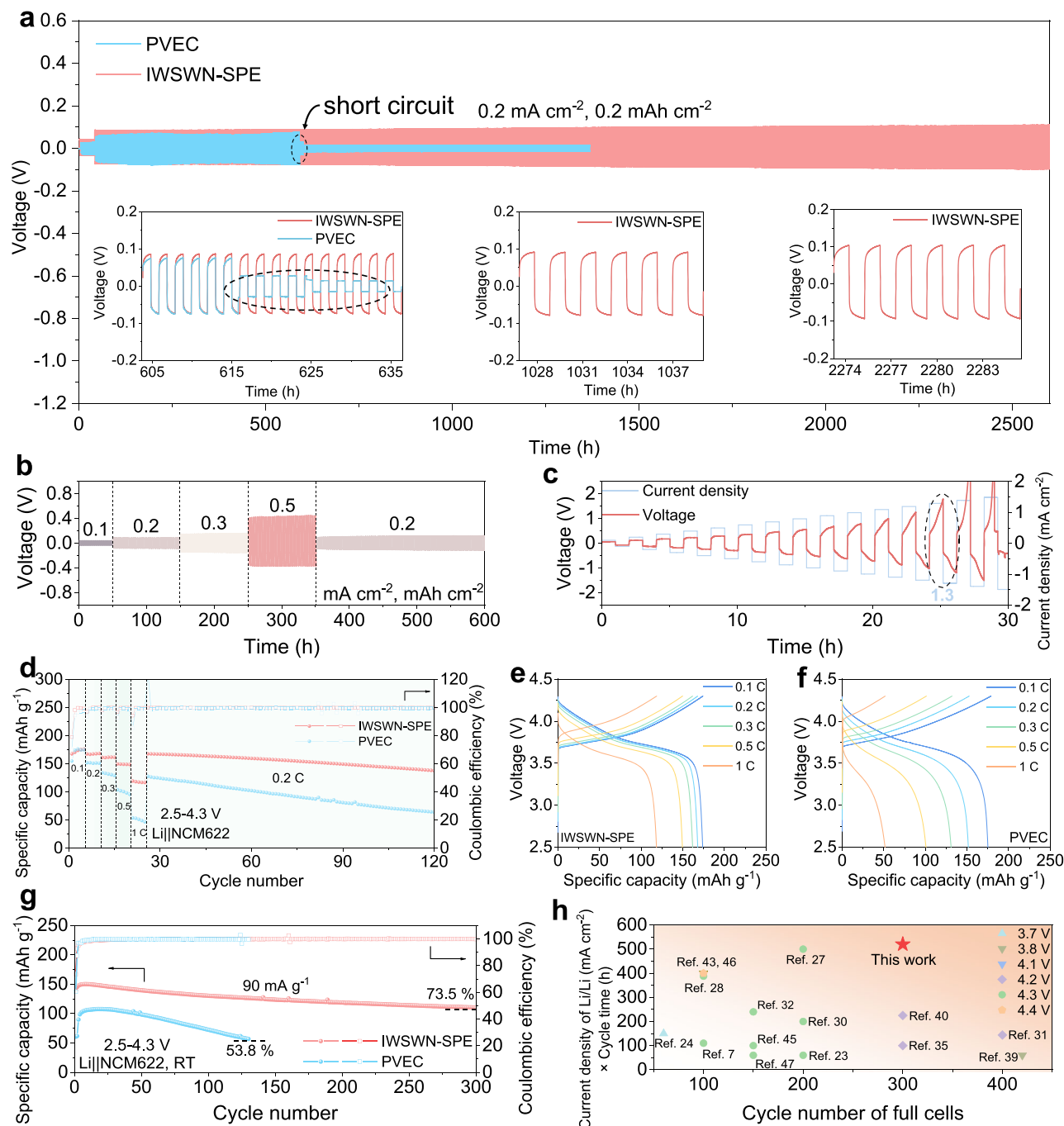


Fig. 4 | The electrochemical performance of batteries with PVEC and IWSWN-SPE at 30 °C. a Long-term cycling of symmetrical Li||Li cells using the PVEC and IWSWN-SPE. **b** Rate performance of Li||IWSWN-SPE|Li cell. **c** Critical current density test of Li||IWSWN-SPE|Li cells. **d** Rate performances of Li||NCM622 cells using PVEC and IWSWN-SPE. (1 C = 180 mA g⁻¹). **e** Charge-discharge curves of Li|PVEC|NCM622

cells at different rates. **f** Charge-discharge curves of Li|IWSWN-SPE|NCM622 cells at different rates. **g** Cycle performances of the Li|PVEC|NCM622 and Li|IWSWN-SPE|NCM622 cells at 0.5 C. **h** Comparison of cycling performance of different polymer-based LMBs.

0.5C. The galvanostatic charge/discharge voltage curves during the cycling are shown in Figs. S25 and S26. At the initial ten cycles, the specific discharge capacity of the Li|IWSWN-SPE|NCM622 cell reaches a maximum value of 150 mAh g⁻¹ and retains 73.5% after 300 cycles. In contrast, the Li|NCM622 cell with the PVEC can only maintain 53.8% of its maximum capacity of 107.8 mAh g⁻¹ after 130 cycles.

The above results demonstrate that the introduction of TAP and HFBMA molecule fragments can significantly improve the cycling performance of PVEC in LMBs, paring with a high-voltage NCM622 positive electrode. The comparisons with the recently reported

polymer-based solid-state electrolytes in terms of their symmetric Li||Li and Li||positive electrode cycling performances are illustrated in Fig. 4h and Table S3^{7,23,24,27,28,30–32,35,39,40,43,47–49}. To compare the LMAs' performance, we use the product of cycle time and current density as a comparative standard, listed on the y-axis, which aids in a standardized comparison of lithium deposition capacity. We compare the cycle number and charge cut-off voltage on the positive electrode side performance, enabling a multidimensional comparison. The performance of our IWSWN-SPE excels in terms of both compatibility with LMAs and durability with high-voltage positive electrodes.

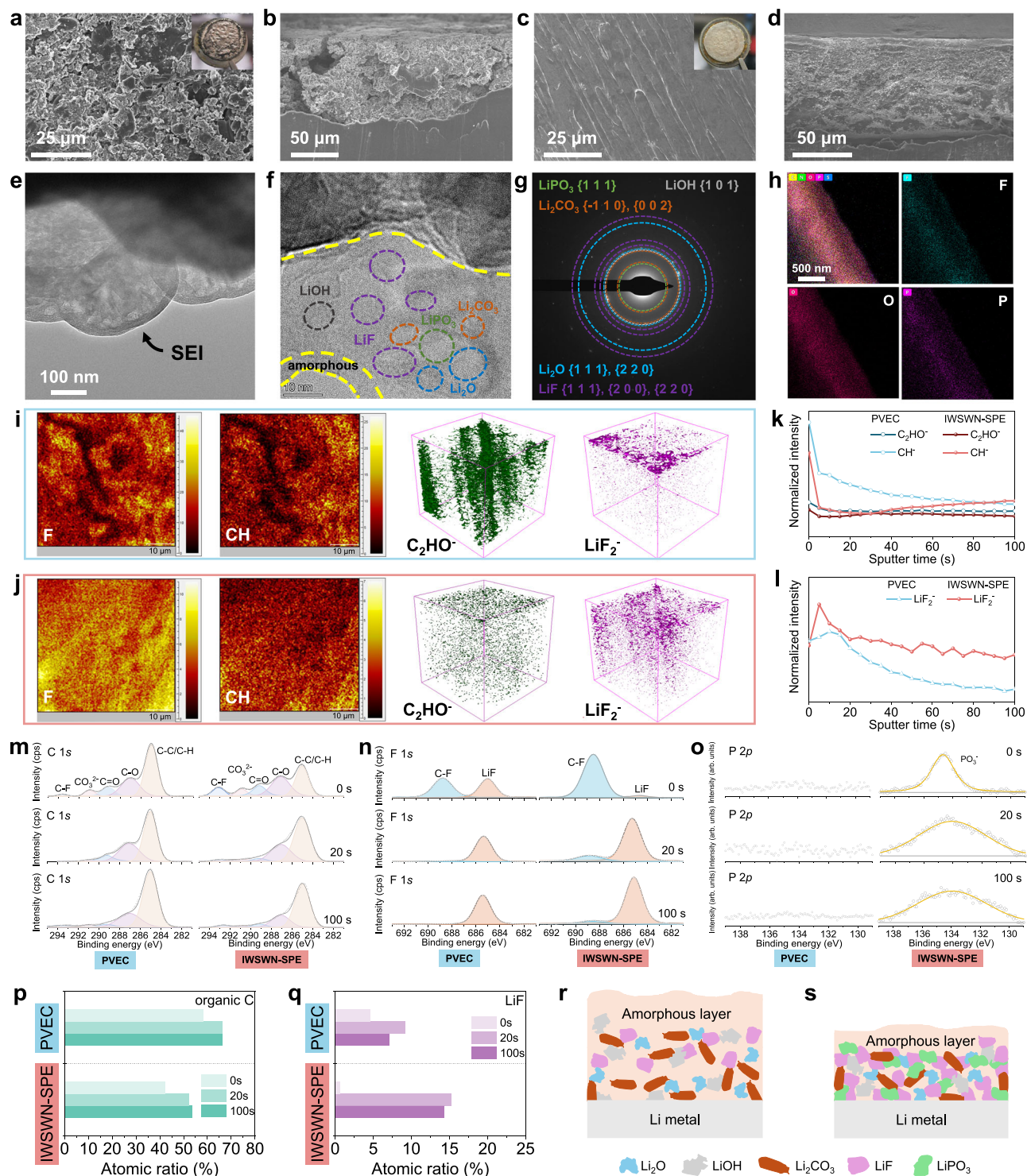


Fig. 5 | Characterizations of SEIs. **a, b** SEM images of the surface (**a**) and cross-section (**b**) of cycled LMAs collected from symmetrical Li||Li cells using PVEC. **c, d** SEM images of the surface (**c**) and cross-section (**d**) cycled LMAs from Li||IWSWN-SPE||Li cells. **e, f** Cryo-TEM images of deposited Li in the cell with the IWSWN-SPE. **g** Corresponding FFT pattern of the inner SEI. **h** EDS mappings of the surface of the deposited Li. **i, j** TOF-SIMS mappings of the LMAs in cycled Li||Li cells

with the PVEC (**i**) and the IWSWN-SPE (**j**). **k, l** Depth profiles of the LMAs in cycled Li||Li cells with the PVEC and IWSWN-SPE. **m–o** C 1s (**m**), F 1s (**n**), and P 2p (**o**) XPS spectra of the cycled LMAs in PVEC and IWSWN-SPE. **p, q** Organic C (**p**) and LiF (**q**) contents derived from XPS results of the cycled LMAs with PVEC and IWSWN-SPE. **r, s** Schematic of the different structures of SEI in PVEC (**r**) and IWSWN-SPE (**s**).

Characterizations of SEIs

To gain a comprehensive understanding of the underlying mechanism of the improved electrochemical stability of LMA in cells with IWSWN-SPE, the surface and cross-section morphology of the cycled LMAs are investigated. Fig. 5a, b show the disassembled Li foil after Li plating/

stripping in the PVEC, revealing an uneven surface and a porous cross-section characterized by extensive mossy-like aggregations and cracks. These observations indicate the occurrence of severe interfacial reactions. In contrast, a compact and homogeneous surface and a dense cross-sectional morphology are showcased for the cycled Li

foil in the IWSWN-SPE, indicating the favorable Li plating/stripping behaviors (Fig. 5c, d). The structure and composition of SEI on the LMAs are studied using cryo-transmission electron microscopy (cryo-TEM)⁵⁰. A continuous and uniform SEI is found on the surface of the deposited Li from the cell with IWSWN-SPE (Fig. 5e). In contrast, an inhomogeneous SEI is formed on the Li for the cell with PVEC (Fig. S27). Moreover, a dual-layered SEI with an inorganic inner phase and an amorphous outer layer is observed in the IWSWN-SPE system (Fig. 5f). The inner inorganic-rich layer consists primarily of LiF, along with smaller quantities of Li_2CO_3 , LiOH, LiPO_3 , and Li_2O species. Characteristic bright diffraction spots corresponding to the LiF (111) and (200) planes are captured in the fast Fourier transform (FFT) pattern (Fig. 5g). EDS mapping tests are also performed to characterize the elemental distribution at the Li interface. As shown in Fig. 5h, it can be observed that F, O, and P elements are uniformly distributed on the interface layer of Li.

The time-of-flight secondary ion mass spectroscopy (TOF-SIMS) is utilized to further analyze the components and structure of the SEI. Fig. 5i, j show the 2D and 3D rendering patterns of the TOF-SIMS and top-down depth sputtering figures, demonstrating that the distribution of various organic and inorganic components on the Li cycled in the IWSWN-SPE is more uniform than that in PVEC. For the cycled LMA using PVEC, C_2HO^- is distributed throughout the entire SEI layer, while LiF_2^- is less present in the internal layers. This indicates that the SEI formed with PVEC is organic-rich. For the IWSWN-SPE, C_2HO^- is primarily concentrated in the surface layer, while LiF_2^- is distributed throughout the SEI layer, suggesting that the SEI formed in IWSWN-SPE is more inorganic-rich. Fig. 5k, l further demonstrate the TOF-SIMS depth sputtering figures of SEI on LMAs. The outer layer of the SEI is abundant with CH^- and C_2HO^- species (the content decreases with the etching depth increases), in which the amount using the PVEC is much more than that using the IWSWN-SPE. For the inner layer, the amount of LiF_2^- species using the PVEC is much less than that using the IWSWN-SPE, showing that the formed SEI using the PVEC has less LiF. Also, PVEC exhibits the peak of LiF_2^- content at a later stage compared to IWSWN-SPE, indicating that the formed outer layer of organics using the PVEC is thicker. Additionally, P-containing inorganic content can be observed in the SEI when using the IWSWN-SPE (Fig. S28), which contributes to facilitating Li^+ transport⁵¹.

XPS depth profiling is also employed to detect the SEI compositions formed in both electrolytes (Fig. 5m–o). In the C 1s spectra, both electrolytes exhibit species such as C-C/C-H, C-O, C=O, CO_3^{2-} , and C-F. However, the formed SEI using the PVEC shows a higher concentration of C-C/C-H species, whereas the formed SEI using the IWSWN-SPE contains a higher proportion of C-F components. This is due to the involvement of the introduced F-containing HFBMA fragments in the composition of the SEI. The organic C content under etching for 0, 20, and 100 seconds is 42.3%, 52.1%, and 53.4%, respectively. These values are lower compared to the SEI formed using PVEC, which has organic C contents of 58.4%, 66.4%, and 66.2% for the same etching times (Fig. 5p). This indicates that the SEI formed in the PVEC is richer in organic content.

In the F 1s spectra, it can be observed that the surface of SEI derived from IWSWN-SPE has more C-F components compared to that derived from PVEC, indicating that the introduced HFBMA fragments participate in the reduction reaction on the LMA. The LiF content in the SEI formed in the IWSWN-SPE is 15.3% and 14.4% after etching for 20 s and 100 s, respectively, whereas the LiF content from PVEC is 9.23% and 7.07%, respectively. The results confirm that the SEI using the IWSWN-SPE has a higher abundance of inorganic LiF content (Fig. 5q). Based on the spectra of P 2p, it can be observed that the introduction of P-containing TAP molecule segments can form an inorganic component LiPO_3 in the composition of the SEI, which is consistent with the results of TOF-SIMS. Fig. 5r, s illustrate the SEI structure on the LMA using different polymer electrolytes. Compared to PVEC, the dual-layer

SEI formed with the IWSWN-SPE features a thinner amorphous organic outer layer and a more substantial inorganic inner layer. This configuration more effectively suppresses Li dendrite growth, reduces Li inventory losses, and enhances the long-term Li plating/stripping performance of LMAs.

Characterizations of CEIs

The surface phase transition from a layered to resistive rock-salt NiO-like structure is known to degrade the NCM positive electrode performance⁵⁰. The cycled NCM622 positive electrodes disassemble from the Li||NCM622 cells with different polymer electrolytes are investigated. Figure 6a, b show that the NCM622 positive electrode cycled in the PVEC exhibits significant structural cracks and damage, which exposes additional fresh surfaces to the electrolyte, leading to extensive formation of the CEI. Therefore, a thick (~7 nm) and uneven CEI is formed on the NCM622 positive electrode (Fig. 6e). Such CEI causes positive electrode surface cracking, electrolyte penetration, and CEI reconstruction, which in turn induces continuous side reactions and rapid capacity decay⁵². In comparison, the NCM622 positive electrode cycled in the IWSWN-SPE maintains excellent structural integrity without visible cracks (Fig. 6c, d), and a thin and uniform CEI layer (~2 nm) is seamlessly wrapped on the NCM622 surface (Fig. 6f). Furthermore, the NCM622 particle cycled in the IWSWN-SPE retains its layered structure, with a smaller rock-salt thickness compared to the NCM622 cycled in the PVEC (Fig. 6e and Fig. S29).

To characterize the chemical states and components of the CEIs on the NCM622 positive electrodes, XPS characterizations are also conducted (Fig. 6h–j and Fig. S30). The CEI derived from IWSWN-SPE shows a stronger peak of LiF compared with PVEC, and the content of LiF in inner layers exceeds 22.0%, while that of PVEC is only 7.8% (Fig. 6h, k). Additionally, the CEI generated in IWSWN-SPE exhibits a lower organic C atomic concentration than the PVEC (Fig. 6i, l), indicating fewer organic species generated by polymer electrolyte decompositions. A significant distinction lies in the fact that the organic C content in the inner layer of the CEI generated in the IWSWN-SPE is only 29.4%, whereas the corresponding value of CEI formed in PVEC reaches up to 65.1%. It is worth noting that the signal of C-F in the CEI derived from the IWSWN-SPE is significantly stronger, demonstrating that the introduced F-containing HFBMA molecule fragments are involved in the composition of the CEI. Moreover, the XPS peak of O 1s located at 529.7 eV is attributed to the signal of lattice O from NCM622⁵². The signal of lattice O for the NCM622 cycled in PVEC can be easily detected, while the much weaker signal of lattice O of the cycled NCM622 in the IWSWN-SPE is observed (Fig. 6j), indicating that the CEI formed on the NCM622 in PVEC is incomplete and can hardly inhibit the transition metal (TM) dissolution. Based on the results of P 2p spectra (Fig. S30), it is observed that the peak of PO_3^- appears in the IWSWN-SPE, indicating that the P-containing TAP molecule segments participate in the formation of P-based inorganic components in the CEI. These components have the potential to mitigate phase transitions in NCM positive electrodes and enhance Li^+ diffusion^{53,54}.

The TOF-SIMS is performed to further confirm the components and distribution of CEI. The C_2HO^- fragment represents the organic components generated from the polymer electrolytes, while LiF_2^- and $\text{PO}_2^-/\text{PO}_3^-$ fragments represent the inorganic species of LiF and phosphate, respectively. The CEI generated in the IWSWN-SPE presents a much lower signal intensity of C_2HO^- and a higher and more uniform signal intensity of LiF_2^- compared to the PVEC (Fig. 6m, n). Moreover, the formed CEI in the IWSWN-SPE shows a uniform distribution of PO_2^- and PO_3^- species (Fig. 6m and Fig. S31). Furthermore, as shown in Fig. 6o, the signal of NiF_3^- is more prominent in the CEI derived from PVEC compared to that from IWSWN-SPE, indicating that the IWSWN-SPE effectively inhibits transition metal dissolution. It follows that, compared to the PVEC, the thin and homogeneous CEI formed on the NCM622 surface in the IWSWN-SPE not only acts as an effective barrier

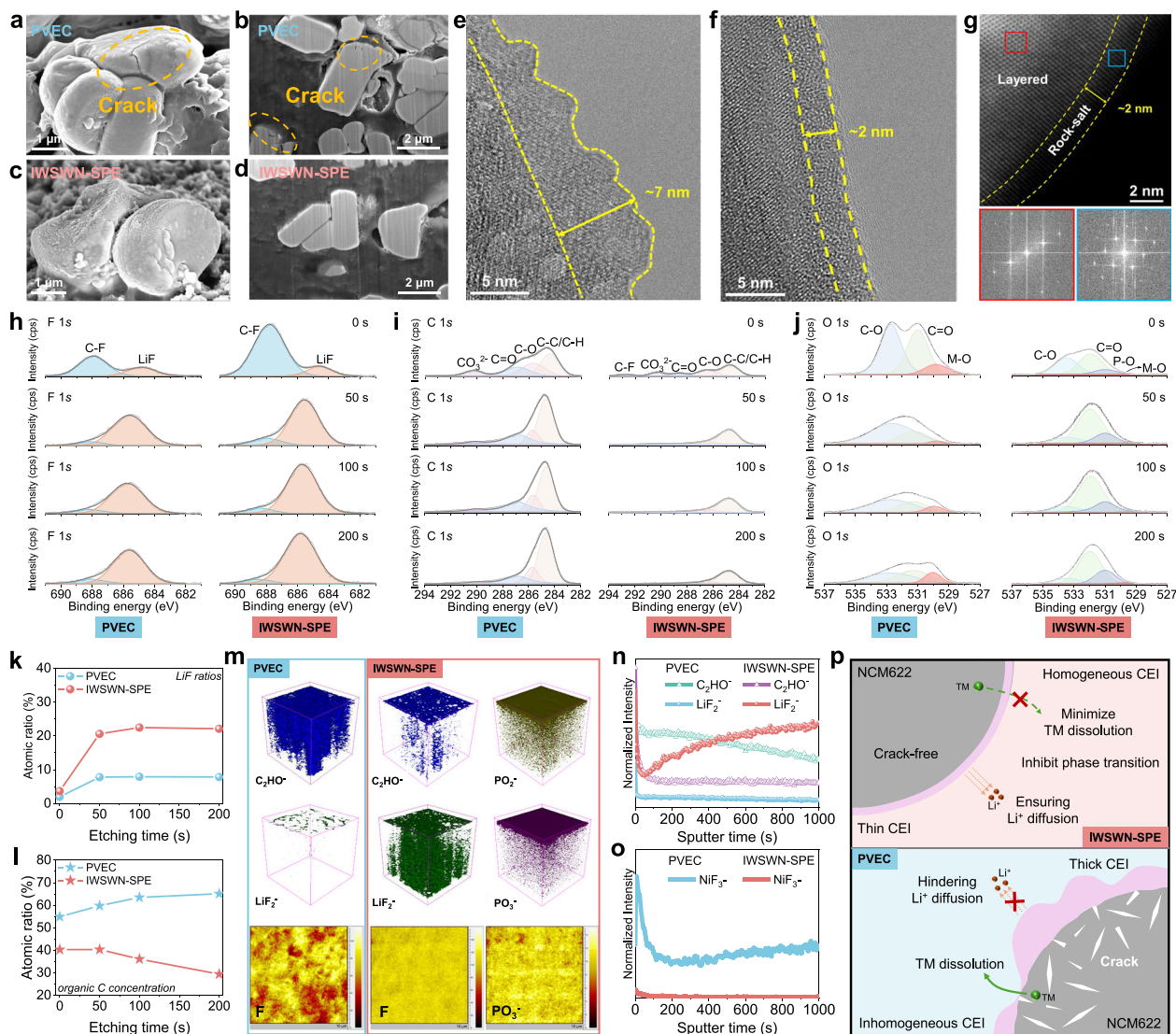


Fig. 6 | Characterizations of CEIs. **a–f** SEM, focused ion beam-SEM (FIB-SEM), and TEM images of the cycled NCM622 positive electrodes with **(a, b, and e)** PVEC and **(c, d, and f)** IWSWN-SPE. **g** High-resolution TEM images and corresponding FFT for the NCM622 particles cycled in the IWSWN-SPE. **h–j** F 1s (**h**), C 1s (**i**), and O 1s (**j**) depth sputtering XPS spectra of the cycled NCM622 positive electrodes in the PVEC and IWSWN-SPE. **k, l** LiF (**k**) and organic C (**l**) content distribution derived from XPS

results of the cycled NCM622 positive electrodes in the PVEC and IWSWN-SPE. **m** TOF-SIMS mappings of the cycled NCM622 positive electrodes in the PVEC and IWSWN-SPE. **n, o** Depth profiles of various elemental segments of the cycled NCM622 positive electrodes in the PVEC and IWSWN-SPE. **p** Schematic showing the structure and mechanism of different CEIs formed in the PVEC and IWSWN-SPE.

against electrolyte side reactions and minimizes TM dissolution, but also mitigates interfacial structural degradation and the formation of micro-cracks within the particles (Fig. 6p)⁵⁵.

Practical applications and safety performance of the IWSWN-SPE

The practicability of the IWSWN-SPE in high-voltage LMBs is demonstrated by testing the performance of the Li||NCM622 cells with a high mass loading of positive electrodes. As shown in Fig. 7a, the Li||NCM622 full cell with a high positive electrode mass loading of 15.8 mg cm⁻² can steadily operate for more than 100 cycles at a cut-off voltage of 4.3 V (with the voltage profiles shown in Fig. 7b). The positive electrode mass loading in our solid-state LMBs based on the IWSWN-SPE is among the highest reported for the polymer-based LMBs to date (Fig. 7c and Table S4)^{7,23,24,27,28,31–33,35,36,39,40,43,47–49}. To further verify the practical feasibility of the IWSWN-SPE, the solid-state Li (50 μm)||NCM622 pouch cell with a high positive electrode loading 12.0 mg cm⁻² is assembled (Fig. S32). The Li||NCM622 pouch cell can

stably cycle for over 30 cycles with a cut-off voltage of 4.3 V at 30 °C without external pressure (Fig. 7d and Fig. S33). Impressively, we further assemble a 2.6-Ah Li||NCM811 pouch cell that achieves a high specific energy of 349 Wh kg⁻¹ based on the 28.2 g of the pouch cell (Fig. 7e, Fig. S34, and Table S5). The results highlight the exceptional Li compatibility and antioxidation properties of our IWSWN-SPE, demonstrating the high potential for use in high-energy-density solid-state polymer LMBs.

Safety performance is another key focus of this study. We investigate the cycling performance of the battery based on the IWSWN-SPE at high temperatures. As shown in Fig. 7f and Fig. S35, even at an elevated temperature of 100 °C, the Li||LFP cell using our IWSWN-SPE exhibits stable cycling for 120 cycles. In contrast, the Li||LFP cell using the PVEC fails after only two cycles. The results indicate that our IWSWN-SPE possesses excellent stability at high temperatures, significantly enhancing the thermal safety of the battery. We also conduct safety testing on the pouch cell. Even exposed to repeated twisting and cutting, the Li||IWSWN-SPE||NCM622 pouch cell continues to power the

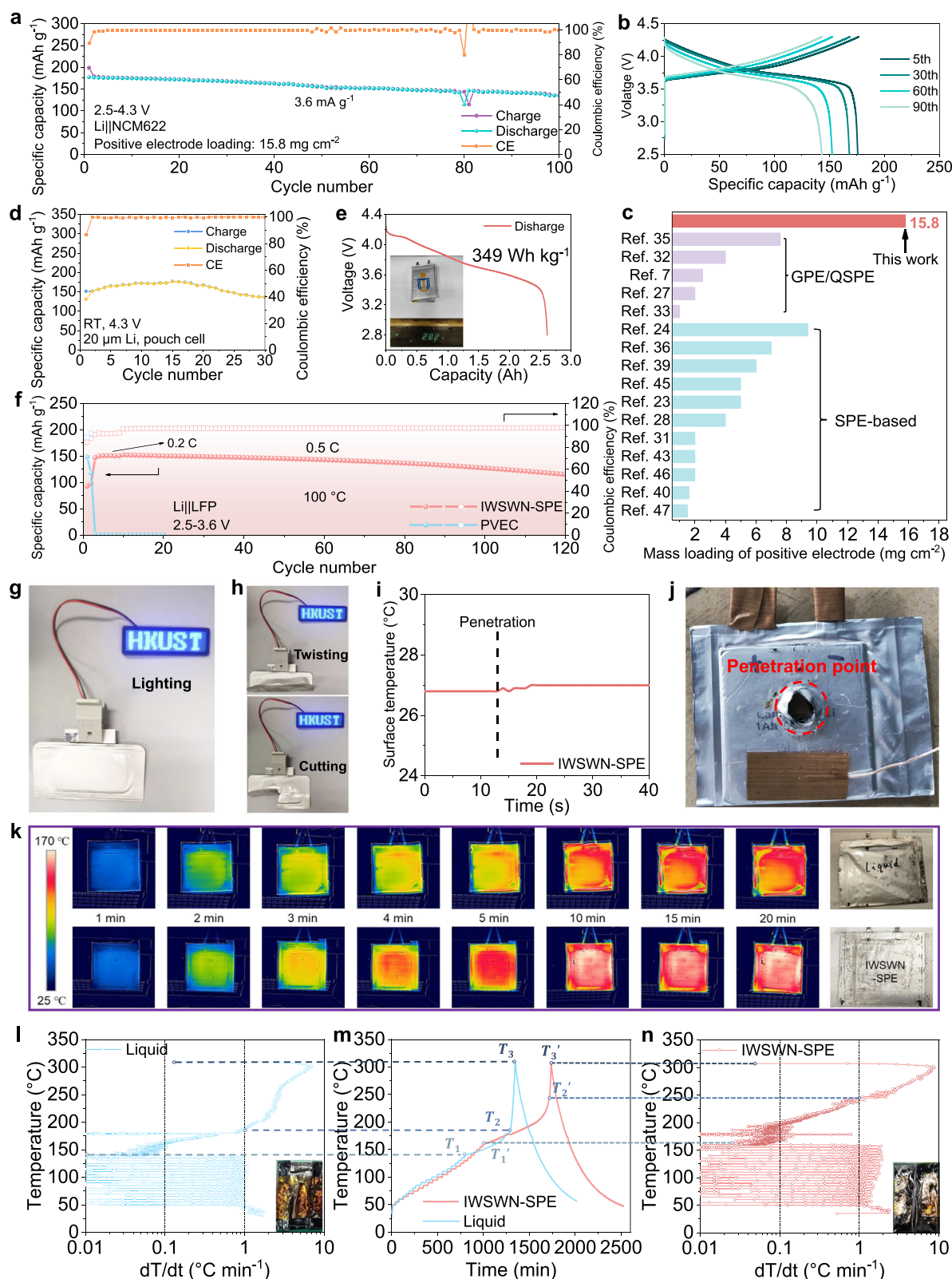


Fig. 7 | Practical applications and safety performance of the IWSWN-SPE.

a, b Cycling performance (**a**) and corresponding voltage profiles (**b**) of the Li||NCM622 cell with a high positive electrode loading of 15.8 mg cm^{-2} operated from 2.5 to 4.3 V. **c** Comparison of the positive electrode mass loading for different polymer-based LMBs. **d** Cycling performance of the Li||NCM622 pouch cell. **e** Discharge profile of the 2.6-Ah solid-state Li||NCM811 pouch cell. **f** Cycling performances of the Li||LFP cells with PVEC and IWSWN-SPE at 100 °C. (1 C = 160 mA g^{-1}). **g, h** Digital

images of LED lighting tests for the Li||IWSWN-SPE||NCM622 pouch cell under rest, twisting, and cutting. **i, j** The nail penetration and the corresponding surface temperature variations (**i**) of 1.0-Ah Li||IWSWN-SPE||NCM811 pouch cell. **k** Evolution of infrared images of the Li||NCM622 pouch cells in the heating plate experiment using commercial liquid electrolyte and IWSWN-SPE. **l–n** ARC tests of the Li||NCM622 pouch cells using commercial liquid electrolyte (**l**) and IWSWN-SPE (**n**).

LED without any signs of liquid leakage, smoke, sparks, or explosions (Fig. 7g, h). To further assess the safety performance of IWSWN-SPE in the battery system with a high-nickel NCM positive electrode, a Li||NCM811 pouch cell is assembled and charged to 4.3 V (1 Ah) for the nail puncture test. As depicted in Figs. 7i, j, and S36, the pouch cell successfully passes the nail puncture test, exhibiting no signs of smog, fire, or combustion. After the penetration, the surface temperature shows a minimal increase without incurrence of thermal runaway.

In addition, we perform heating experiments on the single-layer Li||NCM622 pouch cells in direct contact with a metal heating plate (Fig. 7k). After heating for more than 5 minutes, the Li||NCM622 pouch cell based on the commercial liquid electrolyte (1 M LiPF₆ in EC/DEC with 5 wt% FEC) exhibits uneven heating distribution and swelling. Impressively, when using our IWSWN-SPE in the Li||NCM622 pouch cell, there is uniform heating distribution without any swelling, indicating its superior thermal stability compared to commercial liquid electrolyte. ARC tests are further conducted to compare the thermal stability of the commercial liquid electrolyte and IWSWN-SPE in the Li||NCM622 pouch cells. The three characteristic temperatures [T_1 , T_2 , T_3] are considered crucial indicators of battery thermal runaway. T_1 signifies the point at which the loss of thermal stability, T_2 represents the triggering temperature, and T_3 corresponds to the maximum temperature that a cell can reach during the thermal runaway^{56–58}. As shown in Fig. 7l–n, the T_1 (139.7 °C) of the liquid electrolyte cell is lower than that (160.8 °C) of the IWSWN-SPE, indicating that under the same thermal abuse conditions, our IWSWN-SPE cell is less likely to reach thermal instability temperatures. Furthermore, the triggering temperature T_2 for thermal runaway in the cell using our IWSWN-SPE (244.4 °C) is 55.5 °C higher compared to the cell with the liquid electrolyte, demonstrating that our designed IWSWN-SPE improves the safety and reliability of the batteries.

In summary, we employ a puzzle-like assembly method to construct a high-performance SPE by splicing three distinct puzzle fragments with specific functionality at a molecular level. Specifically, the incorporation of TAP and HFBMA into the VEC matrix enhances the attraction towards TFSI anion and facilitates the dissociation of LiTFSI, resulting in a high Li⁺ transference number of 0.70 and a high ionic conductivity of 0.432 mS cm^{−1} at 25 °C. The molecule segments with intrinsic high-voltage characteristic and the formation of a robust inorganic-rich interfacial layer derived from incorporated segments enable the SPE to withstand oxidation voltages up to 5.15 V. Additionally, the developed SPE exhibits excellent thermal stability and superior nonflammability, enabling stable cycling of Li||LFP cell for over 120 cycles at 100 °C. The solid-state Li||NCM622 cell remains stable through 300 cycles and reliably operates for over 100 cycles even with a mass loading of 15.8 mg cm^{−2}. Notably, a 2.6-Ah solid-state Li||NCM811 pouch cell is assembled and achieves a high specific energy of 349 Wh kg^{−1} (based on the 28.2 g of the pouch cell). Both ARC and nail penetration tests confirm the excellent safety profile of the cells using our developed SPE. This work provides possibilities for customizing advanced polymer electrolyte systems in high-specific-energy LMBs.

Methods

Materials

Active materials (LiNi_{0.6}Co_{0.2}Mn_{0.2}O₂ (NCM622) or LiFePO₄ powders), Super P, polyvinylidene fluoride (PVDF, MW = 1,000,000), anhydrous N-methyl-2-pyrrolidone (NMP, ≥99%) and carbon-coated aluminum (Al) foil (thickness: 12 μm) were purchased from the Guangdong Canrd New Energy Technology Co., Ltd. Ceramic (aluminum oxide, Al₂O₃)-coated Polyethylene (PE) separator film (thickness: 16 μm; porosity: 44 ± 5%; area density: 13.0 ± 1.0 g m^{−2}) was purchased from MTI Corporation-KJ Group. Li foils (20 or 450 μm) were purchased from China Energy Lithium Co., Ltd. LiTFSI (lithium bis(trifluoromethanesulfonyl)imide), azobisisobutyronitrile (AIBN, ≥98%),

vinyl ethylene carbonate (VEC, 99%), and 2,2,3,3,4,4,4-Heptafluorobutyl methacrylate (HFBMA, 97%) were purchased from Sigma-Aldrich. Triallyl Phosphate (TAP, ≥96%) was purchased from Aladdin. The liquid electrolyte (1 M LiPF₆ in EC/DEC with 5 wt% FEC) was purchased from Dodo Chem.

Preparation of the IWSWN-SPE

The monomers of VEC (1 mL), HFBMA (10% molar ratio of VEC), TAP (15% molar ratio of VEC), and LiTFSI (different ratios of the total mass of all monomers, Fig. 2a) were respectively added to a screw bottle, and the AIBN (1.0 wt%, relative to the sum of VEC, HFBMA, and TAP) was added to the screw bottle after the dissolution of LiTFSI, and then the mixtures were stirring at 65 °C for 2 h as “pre-polymerization”. The sticky prepolymer solution was infiltrated into PE separators (diameter: 19 mm) for further polymerization for 6 h at 70 °C. The polymerization of PVEC was the same as the above from the VEC monomer.

Battery assembly

All 2032-type coin cells were assembled in an argon-filled glovebox, with both moisture and oxygen levels maintained below 0.1 ppm. The NCM622 positive electrodes were fabricated via a doctor-blade casting process, using a slurry composed of NCM622 material, Super P, and PVDF in a weight ratio of 8:1:1. The resulting electrodes were punched into 12 mm diameter disks, with a typical mass loading of approximately 3.0 mg cm^{−2} for electrochemical testing. LFP positive electrodes were prepared using the same procedure as that for NCM622, with comparable mass loading values around 3.0 mg cm^{−2}. During coin cell assembly, 50 μL of the polymer precursor solution was injected into each cell. Unless otherwise specified, all Li metal electrodes used in this work had a thickness of 450 μm and a diameter of 15.6 mm. For the pouch cells, the volume of the precursor solution was 5 mL, and cycling performance was evaluated at a pressure of 250 kPa. All the above-mentioned galvanostatic charge-discharge (GCD) tests were operated using a Neware Battery Test System at 30.0 ± 0.1 °C in the Neware Battery Test Incubator.

Characterizations

After Li plating/stripping for 500 h at 0.2 mA cm^{−2} and 0.2 mAh cm^{−2}, the symmetrical Li batteries were disassembled in an argon-filled glove box. The Li sheet can be tested for surface morphology and chemical composition after treatment with DME. SEM and energy-dispersive spectral mapping images were obtained with a JEOL-7100F instrument. The surface morphology and chemical composition of the Li negative electrode were analyzed by XPS (Kratos Axis Ultra DLD), Time of Flight Secondary Ion Mass Spectrometry (TOF-SIMS), and SEM. Cryo-TEM was used to observe the morphology and composition of the interface at the atomic level. For the characterizations of the NCM622 positive electrodes, the Li||NCM622 batteries were cycled for 100 cycles at 36 mA g^{−1} and disassembled in an argon-filled glove box. TEM and STEM were conducted by a JEM-ARM200F (JEOL) instrument. The focused ion beam (FIB) (FEI Helios G4 UX) was used for NCM622 particle cross-section analysis. AFM images and area potential profile were simultaneously collected using the Bruker PT Scanning Probe Microscope (Dimension ICON, Bruker).

Electrochemical impedance spectroscopy (EIS) measurements were performed using a Biologic SP 300 potentiostat over a frequency range of 7 MHz to 1 Hz, applying a sinusoidal excitation voltage with an amplitude of 10 mV. Stainless steel electrodes were employed as blocking electrodes. The ionic conductivity of the solid-state electrolyte was determined using the following relation (1):

$$\sigma = \frac{L}{RS} \quad (1)$$

where L is the thickness of the solid-state polymer membrane (cm), R is the resistance extracted from the intercept on the real axis of the Nyquist plot (Ω), and S is the contact area of the electrode (cm^2). Activation energy E_a was calculated from the Arrhenius Eq. (2):

$$\sigma = A \exp\left(\frac{E_a}{k_b T}\right) \quad (2)$$

where A stands for the pre-exponential factor, k_b stands for the Boltzmann constant, and T stands for the absolute temperature (K). Direct-current (DC) polarization with AC impedance in a Li||Li symmetrical lithium cell was applied to calculate the lithium transference number t_{Li^+} , which can be calculated by the following Eq. (3):

$$t_{\text{Li}^+} = \frac{I_{\text{ss}}(\Delta V - I_0 R_i^0)}{I_0(\Delta V - I_{\text{ss}} R_i^{\text{ss}})} \quad (3)$$

where ΔV is the applied voltage (10 mV), I_{ss} and I_0 are the steady current and initial current respectively during the DC polarization, R_i^0 and R_i^{ss} are the initial and steady charge-transfer resistances respectively during the DC polarization process. The electrochemical stability window of the electrolyte was evaluated using linear sweep voltammetry (LSV) on Li||Al cells, with the voltage scanned from the open-circuit potential up to 5.3 V at a rate of 1 mV s^{-1} at 25°C . Thermogravimetric analysis (TGA) was conducted on a TGA Instruments Q5000 under an argon atmosphere, with the temperature ramped from 25°C to 800°C at a heating rate of $10^\circ\text{C min}^{-1}$. ^{13}C nuclear magnetic resonance (^{13}C NMR) experiments for all monomers were run on a Bruker AVANCE III 600 MHz NMR spectrometer with dimethylsulfoxide- d_6 (DMSO- d_6) as the solvent. Solid-state ^7Li , ^{19}F and ^{31}P NMR spectra with magic angle spinning of all samples were collected on a Bruker Avance NEO solid-state 500 MHz NMR spectrometer. FTIR spectroscopy tests were carried out on Vertex 70 Hyperion 1000 in ATR mode. The Raman spectroscopy tests were carried out on the Micro Raman/Photoluminescence System (InVia, Renishaw).

Computational methods

The molecular dynamics (MD) simulations in this study were conducted using the GROMACS simulation package (version 2024.2)^{59–61} with the OPLS-AA force field⁶². The parameters and atomic charges of Li^+ as reported by Soetens et al were used⁶³, while the force field parameters of TFSI[−], VEC, PVEC and IWSWN-SPE were obtained from LigParGen^{64,65}. For TFSI[−] anion, the Restricted Electrostatic Potential (RESP) atomic charges were used, while 1.2*CM5 atomic charges were employed in organic molecules. All atomic charges were obtained with the Multiwfn 3.8 program⁶⁶ based on the density functional theory (DFT) calculations at B3LYP/6-311 G(d,p) level in Gaussian 09 software⁶⁷. All simulations used an MD timestep of 2 fs. Particle Mesh Ewald (PME)^{68,69} algorithm was used to compute the short-range electrostatics in 2.0 nm, while the cut-off for van der Waals interactions was set as 2.0 nm. The temperature and pressure were controlled through V-rescale coupling and C-rescale barostat, respectively.

The structures of PVEC and IWSWN-SPE used for MD simulations are shown in Fig. S37. In a typical simulation, the systems were initially equilibrated in the isothermal-isobaric ensemble (NPT) ensemble at 398.15 K and 1.01325 bar for 5 ns, after the energy minimization step, in order to allow the polymer molecules to fully relax. And then the systems were annealed from 398.15 K to 298.15 K with a cooling rate of 10 K ps^{-1} . Subsequently, simulation in the NPT ensemble at 298.15 K was carried out for 50 ns to ensure that the system is fully equilibrated (as shown in Supplementary Data 1–4). Finally, the production runs of three parallel sampling processes in the NPT ensemble at 298.15 K for 30 ns were performed as the end.

The HOMO and LUMO energies were computed using DFT as implemented in the Gaussian 09 software package. The calculations employed the B3LYP functional and the 6-311 G (d, p) basis set, which have been widely used for electronic structure analysis in polymer electrolytes. Based on the optimized molecular structures (refer to Supplementary Data 5–7), the HOMO and LUMO characteristics were further analyzed using the natural bond orbital (NBO) theory to gain deeper insights into the electronic distributions.

Density functional theory (DFT) calculations were performed using the Gaussian 09 package at the B3LYP level with the 6-31++ basis set to evaluate the binding energies (BEs) between selected molecules and $\text{Li}^+/\text{TFSI}^-$ ions. All molecular geometries (refer to Supplementary Data 8–11) were fully optimized until the maximum force component on any atom was reduced below 0.00045 atomic units. The binding energy between two species was calculated using the following expression (4):

$$BE = H_{\text{total}} - \sum H_{\text{frag}} \quad (4)$$

where the H_{total} represents the enthalpy of the combined system, and H_{frag} denotes the enthalpy of the individual fragments. Under this convention, a more negative BE indicates a stronger interaction between the two components.

Data availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Information. Source data are provided with this paper.

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Acknowledgments

The work described in this paper was fully supported by a grant from the Research Grants Council of the Hong Kong Special Administrative Region, China (Project No. R6005-20), Guangdong Basic and Applied Basic Research Foundation (No. 2023B1515120022), Guangdong Major Project of Basic and Applied Basic Research (2023B0303000002) and high level of special funds (G03034K001).

Author contributions

T. Zhao supervised this work J. Chen and Y. Li designed this work. J. Chen carried out the synthesis and the experiments. J. Chen conducted the electrochemical experiments and the material characterizations. J. Chen, C. He, X. Peng, J. Li, X. Xu, Y. Zhou, J. Shen, J. Sun, and Y. Li did the data analysis. J. Chen and Y. Li drew the schematic. J. Chen and Y. Li wrote the paper. T. Zhao and Y. Li revised the manuscript. All authors commented on the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-025-63439-6>.

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Peer review information *Nature Communications* thanks Yue Gao who co-reviewed with Huajing Li and the other anonymous, reviewers for their contribution to the peer review of this work. A peer review file is available.

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